5. The stopping of time at light speed is an interesting notion, but it is important not to read too much into it. Special relativity shows that no material object can ever attain light speed: the faster a material object travels, the harder we'd have to push it to further increase its speed. Just shy of light speed, we'd have to give the object an essentially infinitely hard push for it to go any faster, and that's something we can't ever do. Thus, the "timeless" photon perspective is limited to *massless* objects (of which the photon is an example), and so "timelessness" is permanently beyond what all but a few types of particle species can ever attain. While it is an interesting and fruitful exercise to imagine how the universe would appear when moving at light speed, ultimately we need to focus on perspectives that material objects, such as ourselves, can reach, if we want to draw inferences about how special relativity affects our experiential conception of time.

6. See Abraham Pais, *Subtle Is the Lord*, pp. 113–14.

7. To be more precise, we *define* the water to be spinning if it takes on a concave shape, and not spinning if it doesn't. From a Machian perspective, in an empty universe there is no conception of spinning, so the water's surface would always be flat (or, to avoid issues of the lack of gravity pulling on the water, we can say that the tension on the rope tied between two rocks will always be slack). The statement here is that, by contrast, in special relativity there *is* a notion of spinning, even in an empty universe, so that the water's surface can be concave (and the tension on the rope tied between the rocks can be taut). In this sense, special relativity violates Mach's ideas.

8. Albrecht Fölsing, *Albert Einstein* (New York: Viking Press, 1997), pp. 208–10.

9. The mathematically inclined reader will note that if we choose units so that the speed of light takes the form of one space unit per one time unit (like one light-year per year or one light-second per second, where a light-year is about 6 trillion miles and a light-second is about 186,000 miles), then light moves through spacetime on 45-degree rays (because such diagonal lines are the ones which cover one space unit in one time unit, two space units in two time units, etc.). Since nothing can exceed the speed of light, any material object must cover less distance in space